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Operation Analysis of a One-DOF Pantograph Leg Mechanisms

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Abstract. In this paper, operation analysis of a one degrees of freedom (DOF) pantograph leg mechanisms is carried out. The leg mechanisms consist of a four-bar linkage and a pantograph mechanisms and it is based on a Low-cost Easy-operation idea. Experiment tests of the prototype shows the feasible and drawbacks of the proposed leg mechanisms. Kinematics analysis of the leg mechanisms is carried out. Finally, simulation results reveal the motion characteristics and performance of the leg mechanisms.

Keywords. Biped Robots, Pantograph-leg mechanisms, Mechanism analysis

1. Introduction

It is well known that legged locomotion is more efficient、speedy、and versatile than the one by track and wheeled vehicles when it operates on a rough terrain、steeps stairs or avoids obstacles (Song and Waldron, 1989, Carbone and Ceccarelli, 2005). This research field has attracted great interest in the past few decades and a lot of prototypes have been successfully built in university laboratories or companies. Walking abilities have already achieved for dynamic walking、descending and ascending stairs、passive dynamic walking、running and even jumping. Significant examples can be indicated in HRP-2 developed by AIST, WABIAN-IV humanoid robot in the Takanishi Laboratory, and Honda's ASIMO (Hirai et al., 1998, Yoshiaki et al., 2002, Carbone, and Ceccarelli 2005, Liu et al., 2007).

Most of these robots are aimed to the aspects of dynamic control、 gaits synthesis or human-robot interactive studies. Nevertheless, mechanism design and geometry of biped robots, especially for leg mechanisms, are crucial to build efficient and reliable biped robots. A leg mechanism will not only determine the degrees of freedom of a robot, but also actuation system efficiency and control strategy. Thus, aspects on leg mechanisms are fundamental design and operation issues (Song and Waldron, 1989, Liu et al., 2007).

In order to make a biped robot walking anthropomorphic like, most of the leg mechanisms are build with three or more actuating motors at least with hip、knee and ankle. These kinds of legs have anthropomorphic design and the same degrees of freedom of human legs. Therefore they are anthropomorphic and flexible. However, there are several drawbacks: the design is very complex and expensive for a leg system considering numbers of motors and their proper distribution. Then, because of static unstable characters, sophisticate control algorithms and electronics hardware are necessary. In addition, they are not energy efficient because of the "Back-driven" effect (the force conflict between different motors when a leg swings) and the masses of motors (Song and Waldron, 1989). Therefore, it is very difficulty to apply biped robots walking in a real environment for some tasks.

To make a biped robot more attractive, a different methodology can be considered such as constructing a biped robot with reduced number of degrees of freedom and compact mechanism leg system. Therefore, carefully attention can be paid on a mechanism design of the leg system and actuation system. Characters such as mechanism dimension、workspace、actuating torque should be designed and optimized in depth.

Actually, the reduced DOF of leg mechanisms design is originally from hundreds years ago, for example the wooden horse of Rygg in 1893 (Song and Waldron, 1989). Most of such kinds of leg mechanisms are successfully applied in four-legged or six-legged robots. (Carbone and Ceccarelli, 2005). Several prototypes have validated the effectiveness of reduced DOF leg mechanisms like ASV、OSU、PVII and ODEX (Song and Waldron, 1989). Mechanism design and optimization problems for different kinds of leg mechanisms are studied in some works (Funabashi et al 1991, Shieh et al, 1997). In the works (Takeda et al., 2001, Wu et al., 2007) a walking chair as an alternative vehicle of a wheelchair is studied with leg mechanisms in depth as an application for welfare facilities.

At LARM: Laboratory of Robotics and Mechatronics in Cassino University, research lines are directed to Low-Cost Easy-Operation leg design in order to implement reduced degree of freedom leg mechanisms into a biped robot (Ottaviano et al., 2004). Prototypes have been built in the laboratory LARM and this kind of leg mechanisms has been investigated to be applied even in a low-cost easy-operation humanoid Robot CALUMA (Nava, 2007).

This paper is organized as follows. In the first section the leg mechanisms is introduced as consisting of a four-bar linkages and a pantograph mechanism. In the second section a kinematic description of one leg mechanism is proposed with suitable formulation. In the third section, experiment tests are carried out with a prototype in order to validate the leg mechanism design and to find out drawbacks of the prototype. In the fourth section, simulation is implemented in Matlab simulation environment and simulation results are discussed for performance and characteristics of the proposed leg mechanisms. In the end are the conclusions and future work.

2. Mechanism description

For human normal walking, the motion of a leg can be divided into two phases: a propelling phase and non-propelling phase (Carbone and Ceccarelli, 2005). In Fig. 1, the dashed line represents the supporting leg in propelling phase and the solid line represents the swing leg in non-propelling phase. The motion trajectory of the ankle is depicted with dashed line and it is a symmetric curve. The hip trajectory is also a curve but with little wave. The step length L and step height H are also depicted in Fig. 1.

Therefore, in order to design a leg mechanism with back-forth and up-down motion capability in sagittal plane, the foot point of a reduced DOF leg mechanism should also be able to generate an ovoid curve, which is composed of a straight-line segment and a curved segment. The straight-line segment is related to the propelling phase when the corresponding leg touches the ground and could guarantee stable propelling of the body. The curved segment is related to the non-propelling phase, which is produced by leg when it swings from back to forth.

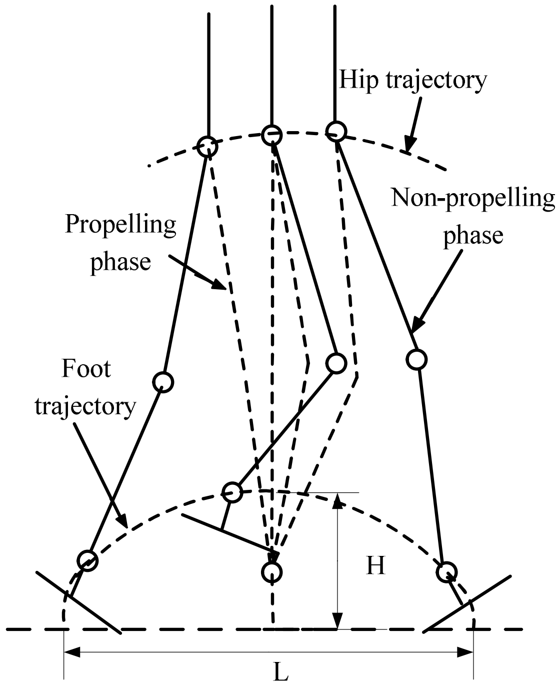


Fig. 1. A scheme for human walking gait

The mechanisms which can produce the straight-line motion are studied intensively in a very rich literature. There are some famous mechanisms, for example the four-bar linkages of Watt and Evans, Richard Roberts' mechanism and Chebyshev linkage (Harteberg and Denavit, 1964). In the works (Shieh et al., 1996) four-bar linkages、six-bar linkages and eight-bar linkages are classified and selected in order to find a proper mechanism structure which is able to produce an ovoid curve with approximate straight-line segments.

In this work, a Chebyshev four-bar linkage design is considered because of its simplicity and characteristics (Mehdigholi and Akbarnejad , 2008). In order to amplify the produced motion of the Chebyshev linkage, a pantograph mechanisms is utilized. Pantograph mechanisms which include the normal and skew configuration in 2D and 3D respectively are used widely in walking mechanisms because of their characteristics (Song and Waldron, 1989).

Therefore, in this paper the proposed one DOF leg mechanisms consists of a Chebyshev four-bar linkage LEDCB and a pantograph mechanism PBHA, as shown with design parameters in the Fig. 2.

The Chebyshev mechanism LEDCB can generate an ovoid curve for the point B. It is also a crank-rocker linkage, where the triangle EDB of coupler is a straight line. The crank is LE, rocker is link CD and coupler triangle is EDB. Joints L、C、P are fixed on the body of the biped robot, and the offset h between them will greatly influence the trajectory shape of point A.

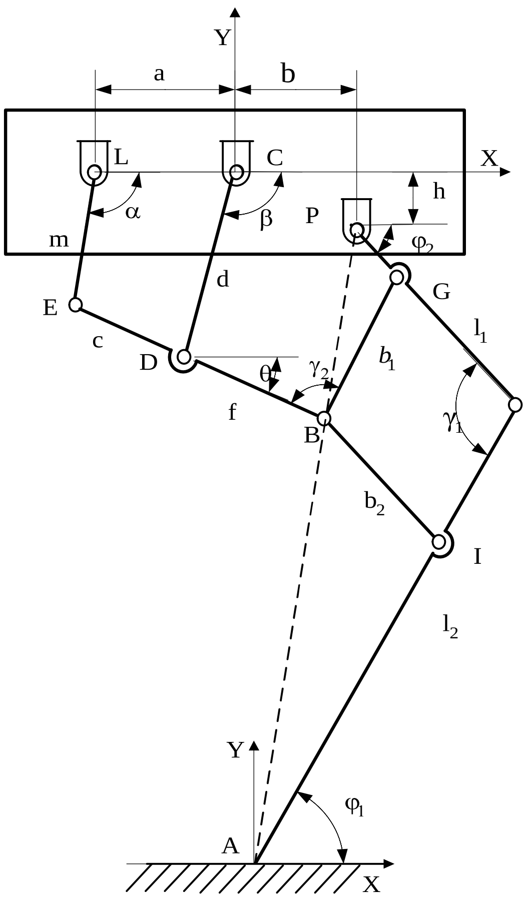


Fig. 2. A scheme for one DOF pantograph leg mechanisms

The pantograph mechanism PBHA is used to amplify the input trajectory of point-B into output trajectory with the same shape at Point-A. The point P is fixed on the body of the robot. The amplify ratio of the pantograph mechanism depends on the length of link H-I and link I-A or the ratio of PA and PB (1-PA/PB). The transmission angles γ_1 and γ_2 are two important parameters for transfer effectiveness of the mechanism. A good performance could be get to keep them as near $\pi/2$ as possible. It is a practical condition that $|\gamma_i - 90^\circ| < 40^\circ \quad (i = 1, 2)$ according to kinematics rules of linkages (Harteberg and Denavit, 1964).

3. A Kinematic analysis

In order to analyse and evaluate the prototype leg mechanisms performance, a kinematic analysis has been carried out.

A coordination system XY is fixed at the point C as the original reference position. Another reference frame XY is fixed on the foot ground point A. Design parameters of the leg mechanism are depicted in Fig. 2. The rotation angle of the crank is α、φ_2 is the angle of the leg with respect to the ground. γ_1 and γ_2 are the transmission angles of the pantograph mechanism. The position of point B can be evaluated as a function of the input crank angle α as (Ottaviano et al., 2004).

$$
\begin{aligned}
x_B &= -a + m\cos\alpha + (c+f)\cos\theta \\
y_B &= -m\sin\alpha - (c+f)\sin\theta
\end{aligned}
\tag{1}
$$

where

$$
\theta = 2\tan^{-1}\left(\frac{\sin\alpha - (\sin^2\alpha + B^2 - D^2)^{1/2}}{B+D}\right) \tag{2}
$$

and

$$
\begin{aligned}
B &= \cos\alpha - \frac{a}{m} \\
C &= \frac{a^2 + m^2 - c^2 + d^2}{2md} - \frac{a}{d}\cos\alpha \\
D &= \frac{a}{c}\cos\alpha - \frac{a^2 + m^2 - c^2 + d^2}{2mc}
\end{aligned}
\tag{3}
$$

By considering the five-bar linkage CDBGP, one can write the closure loop equation to obtain φ_1 and φ_2 as (Ottaviano et al., 2004).

$$
\begin{aligned}
\varphi_1 &= 2\tan^{-1}\frac{1 - \sqrt{1 + k_1^2 - k_2^2}}{k_1 - k_2} \\
\varphi_2 &= 2\tan^{-1}\frac{1 - \sqrt{1 + k_2^2 - k_4^2}}{k_3 - k_4}
\end{aligned}
\tag{4}
$$

where

$$
\begin{aligned}
k_1 &= \frac{x_B - p}{y_B - h} \\
k_2 &= \frac{b_1^2 + y_B^2 + x_B^2 - (l_2 - b_2)^2 + p^2 + h^2 - 2px_B - 2hy_B}{2b_1(y_B - h)} \\
k_3 &= \frac{p - x_B}{y_B - h} \\
k_4 &= \frac{-b_1^2 + y_B^2 + x_B^2 + (l_2 - b_2)^2 + p^2 + h^2 - 2px_B - 2hy_B}{2(l_2 - b_2)(y_B - h)}
\end{aligned}
\tag{5}
$$

Consequently, from Fig. 2, transmission angles γ_1 and γ_2 can be evaluated as $\gamma_1 = \varphi_1 + \varphi_2$ and $\gamma_1 = \pi - \theta - \varphi_1$, respectively.

The position of point A can be given as

$$
\begin{aligned}
x_A &= x_B + b_2\cos\varphi_2 - (l_1 - b_1)\cos\varphi_1 \\
y_A &= y_B - b_2\sin\varphi_2 - (l_1 - b_1)\sin\varphi_1
\end{aligned} \quad (7)
$$

The velocity of point A can be obtained by

$$
\begin{aligned}
\dot{x}_A &= \dot{x}_B - b_2\dot{\varphi}_2\sin\varphi_2 + (l_1 - b_1)\dot{\varphi}_1\sin\varphi_1 \\
\dot{y}_A &= \dot{y}_B - b_2\dot{\varphi}_2\cos\varphi_2 - (l_1 - b_1)\dot{\varphi}_1\cos\varphi_1
\end{aligned} \quad (8)
$$

The acceleration of point A can be obtained as

$$
\begin{aligned}
a_{AX} &= a_{BX} - b_2(\ddot{\varphi}_2\sin\varphi_2 + \dot{\varphi}_2{}^2\cos\varphi_2) \\
&\quad - (l_1 - b_1)(\ddot{\varphi}_1\sin\varphi_1 + \dot{\varphi}_1{}^2\cos\varphi_1) \\
a_{AY} &= a_{BY} - b_2(\ddot{\varphi}_2\cos\varphi_2 - \dot{\varphi}_2{}^2\sin\varphi_2) \\
&\quad - (l_1 - b_1)(\ddot{\varphi}_1\cos\varphi_1 + \dot{\varphi}_1{}^2\sin\varphi_1)
\end{aligned} \quad (9)
$$

Eqs(1) and Eqs(9) can be used for a numerical simulation of the operation of the leg mechanisms with the aim to evaluated and characterize the basic characteristics of the proposed design.

3. Experimental tests

In order to evaluate the feasible and test operation performances of the proposed leg mechanisms, a prototype has been built at LARM in Cassino with mechanism dimension parameters in Tab. 1.

Tab. 1. Design parameters of a prototype leg mechanisms at LARM in Fig. 3

Pantograph mechanisms (mm)		Chebyshev linkage (mm)
a = 50	b = 20	l_1 = 300
h = 30	m = 25	l_2 = 200
c = 62.5	d = 62.5	b_1 = 75
f = 62.5	—	b_2 = 150

Fig. 3 shows the prototype as a biped robot as mainly composed of two one DOF legs. Two big feet are connected to the ankles of the two legs in order to keep balance and enlarge the friction between feet and ground. Legs are connected to the body with a DC motor and flywheels. The actuated crank angles of the two legs are π degree synchronized.

During experiments accelerometers are attached on the output points of the Chebyshev linkage and pantograph mechanism in order to get the acceleration information of leg mechanisms when the prototype operates. The output signals of the accelerometers are acquired through a NI 6024E card installed on a PC. Data are managed and stored in the LabView software environment.

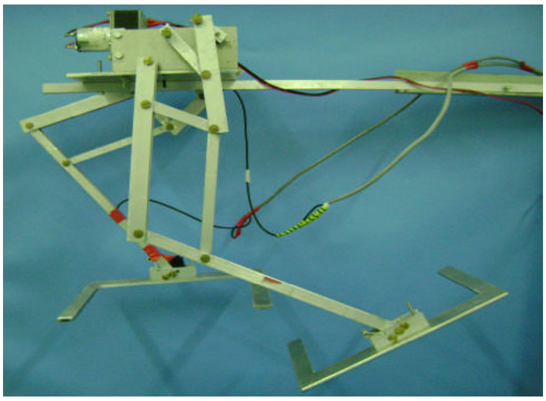


Fig. 3. A prototype of one DOF biped robot built at LARM

The accelerometers on the robot can only measure the acceleration signals along one axis, which is orthogonal the surface of the accelerometer. Therefore, the acquired acceleration signals are not with respect to the global inertial coordinate but are in $X'Y'$. As show in Fig. 4, reference frame $X'Y'$ is fixed on the foot end of the biped robot. Another global inertial coordinator XY is fixed on the ground. Accelerometers can only measure the acceleration signals along the axis X'. In order to get the acceleration values with respect to the inertial coordinates system XY, a coordinate transfer has been computed.

The rotation matrix between XY and $X'Y'$ is

$$
R = \begin{bmatrix} \sin\varphi_1 & -\cos\varphi_1 & 0 \\ \cos\varphi_1 & \sin\varphi_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (11)
$$

Then acceleration along X axis can be expressed as

$$
a'_A = Ra_A \quad g' = Rg \quad a'_A = -a_A - g' \quad (12)
$$

where a'_A and g' represent the foot acceleration and gravity acceleration with respect to reference frame $X'Y'$.

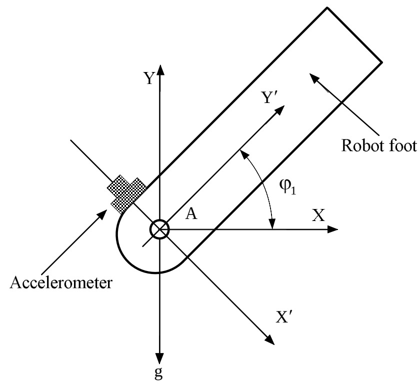


Fig. 4. Position of the accelerometer and coordinate system in the leg in Fig. 3.

In order to know the walking performances, accelerometers are also attached on the body of the robot at point P in order to measure the body acceleration when the biped robot walks on the ground.

Fig. 5 and Fig. 6 show the measured acceleration along X axis and Y axis for point A during lab test when the prototype operates on a supporting stand.

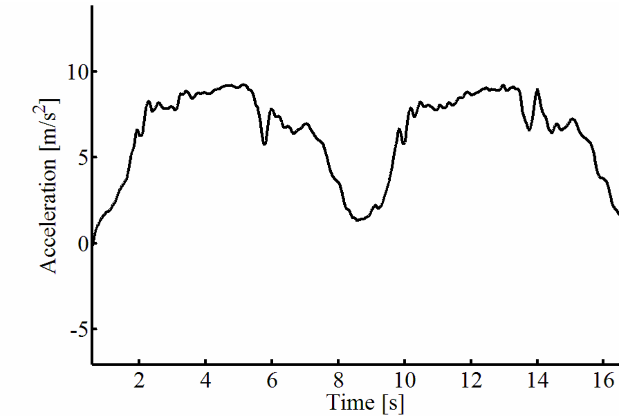


Fig. 5. Measured acceleration along X axis for point A during lab test of the prototype.

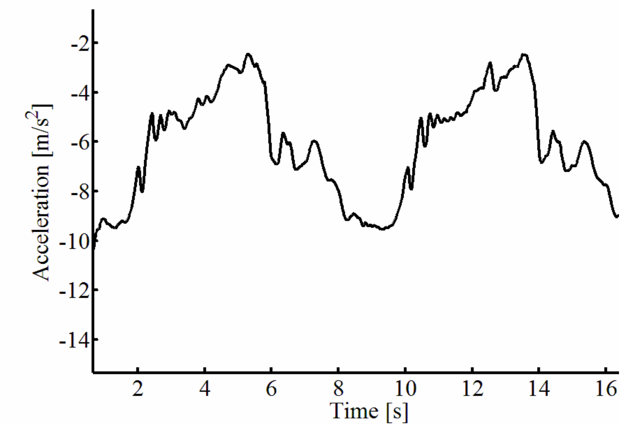


Fig. 6. Measured acceleration along Y axis for point A during lab test of the prototype.

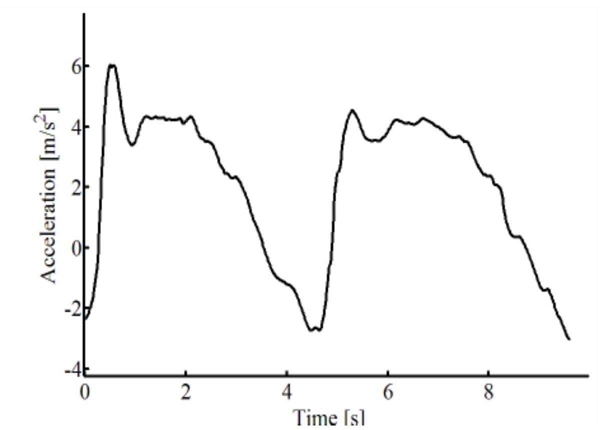


Fig. 7. Measured acceleration along X axis for body point P during lab test of the prototype.

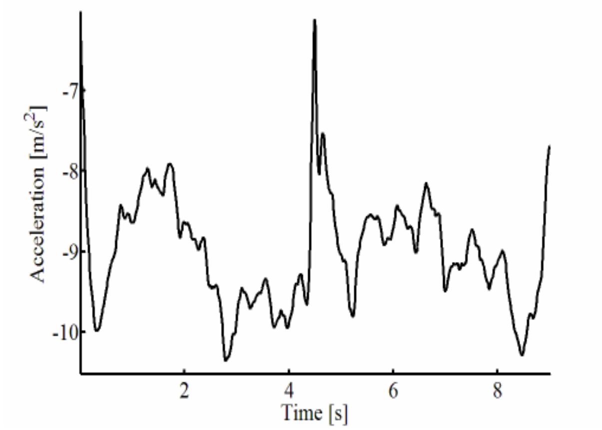


Fig. 8. Measured acceleration along Y axis for body point P during lab test of the prototype.

Fig. 7 and Fig. 8 show the measured acceleration along X axis and Y axis for the robot body.

Fig. 9 shows the snapshots of the walking sequence of the biped robot during a lab test.

As we can see from the experiment results, all the acceleration curves are periodical. The time of one period in Fig. 5 and Fig. 6 are twice as compared with the period in Fig. 7 and Fig. 8 because of symmetric motion of two legs. The plotted results are not smooth curves because of mechanisms vibration and measuring errors.

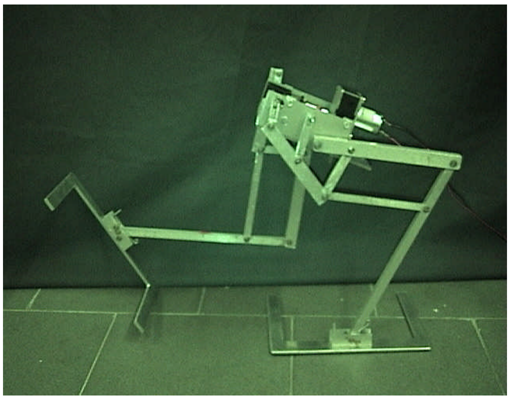
The operation characteristics of the built prototype are shown in the walking sequence in Fig. 9.

One of the two legs touches the ground and it functions as the supporting leg which propels the body of the biped robot forward as show in Fig. 9 (a) and (b). At the same time, the other leg moves with non-propelling role and swings from back to forth. When the swinging leg touches the ground it becomes the supporting leg and go into the propelling phase as shown in Fig. 9 (b) and (c). Then, for each step the phases are repeated sequentially.

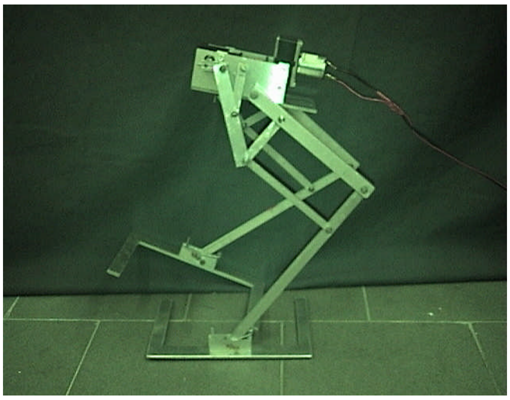
Laboratory experiment results have shown this biped robot dose not walk properly when it walks on the ground. As shown in Fig. 7 the acceleration along X axis is so large that at each step the robot rush forward and its foot hits the ground heavily. Experiment videos show that the biped robot walks just like a “drunk-man”. The walking steps for the biped robot are too large since a step is almost equal to the dimension of the leg mechanisms. Additionally, the mass centre of the biped robot also moves in a large area and sometimes it goes out of the area of balance

4. Simulation Results

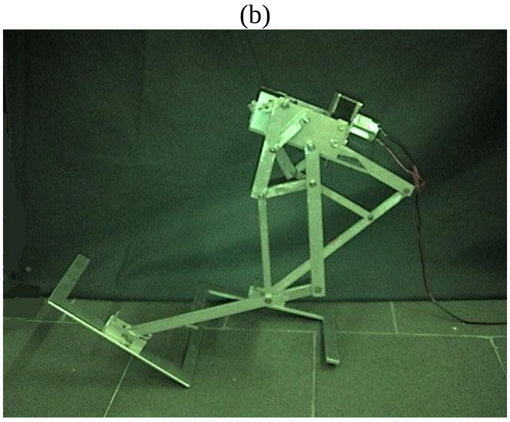
A Matlab simulation program has been developed in order to study the kinematic performance of the proposed leg mechanisms. Simulation parameters of the leg mechanisms are those of the prototype in Table. 1.



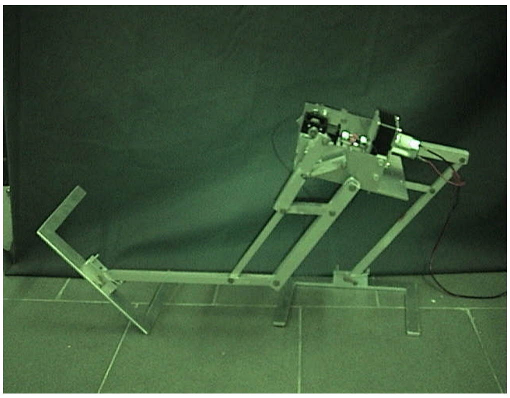
(a)



(b)



(c)



(d)

Fig. 9. The snapshot of the walking sequences by the LARM prototype in a lab test.

The motion trajectories of point A and point B as shown in Fig. 10, when the leg mechanisms operate on a supporting stand as in Fig. 3.

The point B, which is the output-point of the Chebyshev linkage, can trace an ovoid curve with an approximately straight-line and a coupler segment which mimics properly the motion of human ankle. The point A, which is the output-point of the pantograph mechanism can amplify the trajectory of point B very well.

Fig. 11 shows the motion trajectories of point A and Point B in the biped robot when it walks on the ground. The trajectories of point A and point B are depicted with dashed lines. All the trajectories have a short approximately straight line at the beginning and end, because when both legs grasp the ground and it becomes a closed loop mechanism.

Fig. 12 shows the transmission angles of γ_1 and γ_2 with respect to the crank angle α, respectively.

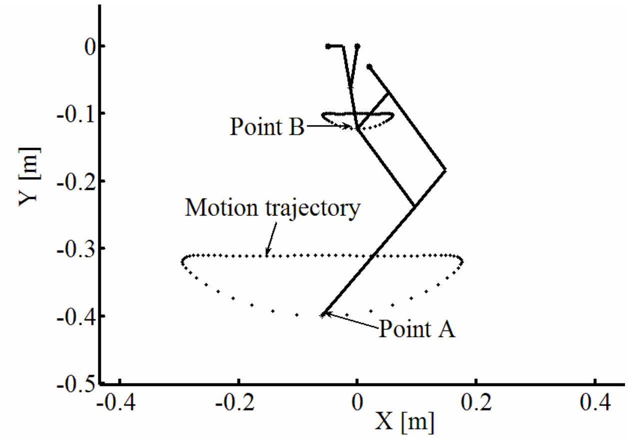


Fig. 10. Simulation results for motion trajectories of point A and B when leg operates on a supporting stand in Fig. 3.

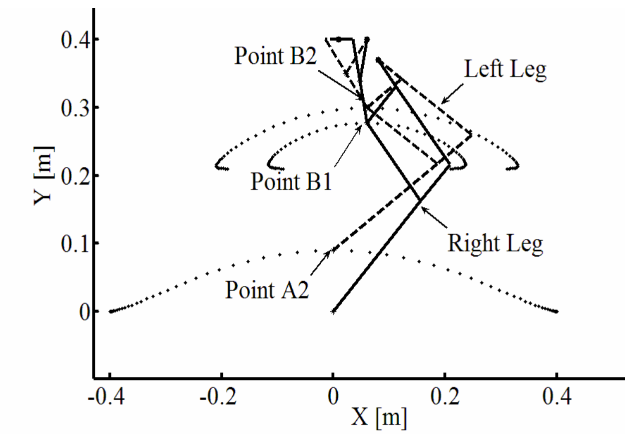


Fig. 11. Simulation results for trajectories of point A and Point B when the biped robot walking on the ground

Thus, An effective transmission of the mechanism could be appreciated since .the transmission angles are between 50 deg and 130 deg.

Fig. 13 and Fig. 14 show the computed acceleration values of point A along X axis and Y axis, respectively.

In order to compare with the simulation results, measured acceleration signals in Fig. 5 and Fig. 6 are also zoomed in. As shown in Fig. 15 and Fig. 16. Comparing Fig. 13 with Fig. 14 and Fig. 15 with Fig 16. One could find out that the experiment results are similar to the simulation results. However, because of the mechanism clearance、vibration, backlash and also measuring errors, experiment results are no coincident with the simulation results. These reflect that there are some design drawbacks in the prototype design.

Fig. 17 shows the computed motion trajectory of the mass center, it is as a periodic arc curve.

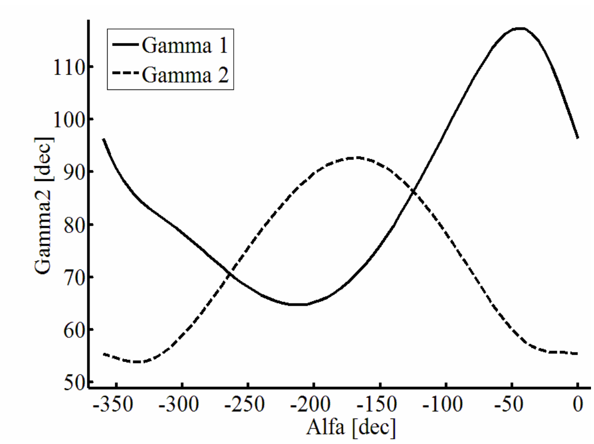


Fig. 12. The transmission angle γ_1 and γ_2 with respect to crank angle α

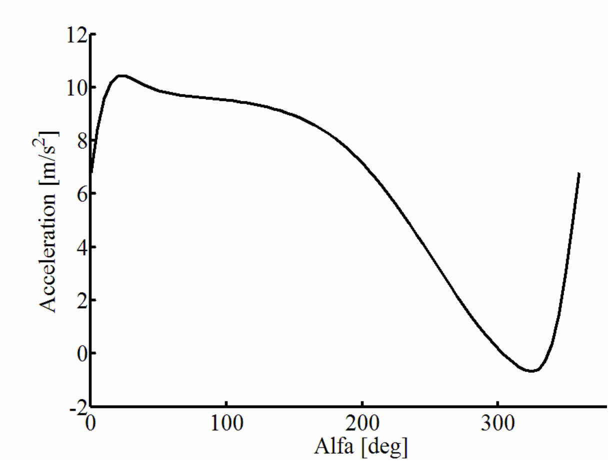


Fig. 13. Computed acceleration along X axis for point A during simulation of one step of walking in Fig 10.

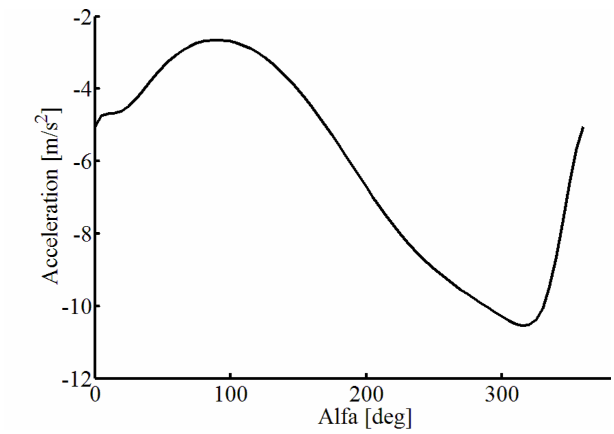


Fig. 14. Computed acceleration along Y axis for point A during simulation of one step of walking in Fig 10.

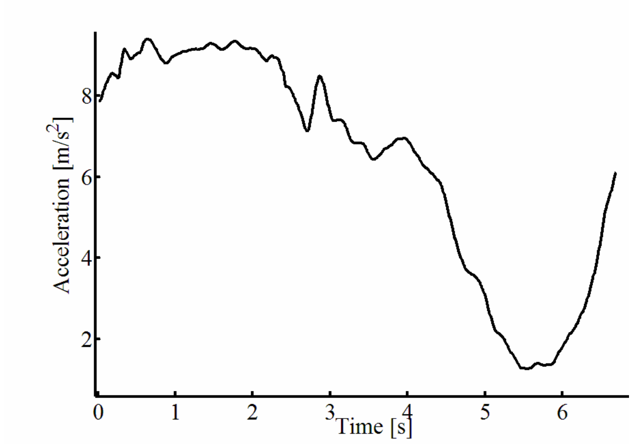


Fig. 15. Measured acceleration along X axis for point A during lab test of the prototype in Fig. 5.

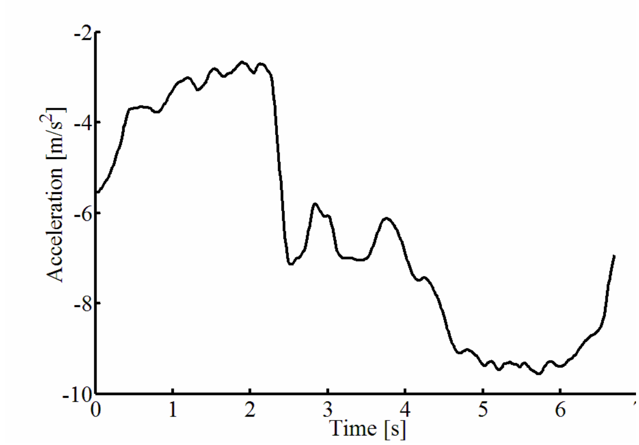


Fig. 16. Measured acceleration along Y axis for point A during lab test of the prototype in Fig. 6.

Fig. 18 shows the computed motion sequence of the biped robot in sagittal plane and Fig. 19 shows the foot motion sequence in horizontal plane.

In Fig. 18, the dashed line depicts the left leg and the solid line depicts the right leg. Two legs grasp the ground alternatively in saggital plane and the walking gaits are produced. The motion trajectories of the feet and point B are also depicted with dashed line.

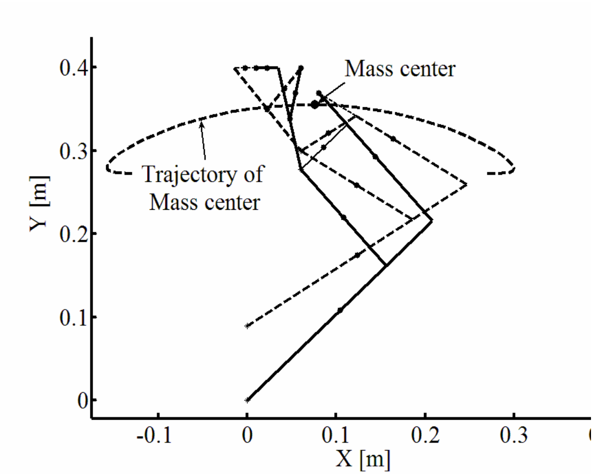


Fig. 17. The motion trajectory of the mass center when biped robot walks on the ground

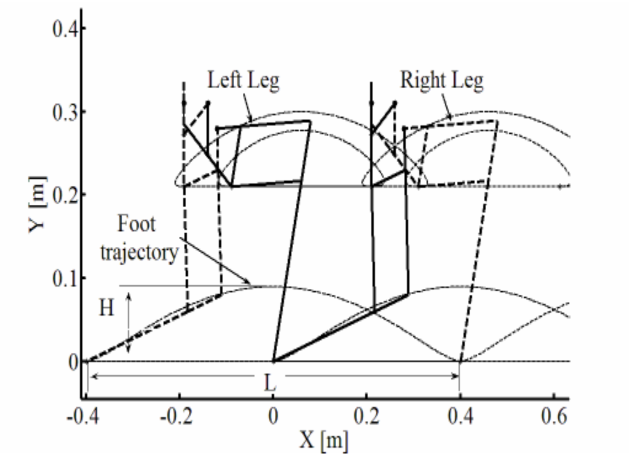


Fig. 18. The motion sequences in sagittal plane when biped robot walks on the ground

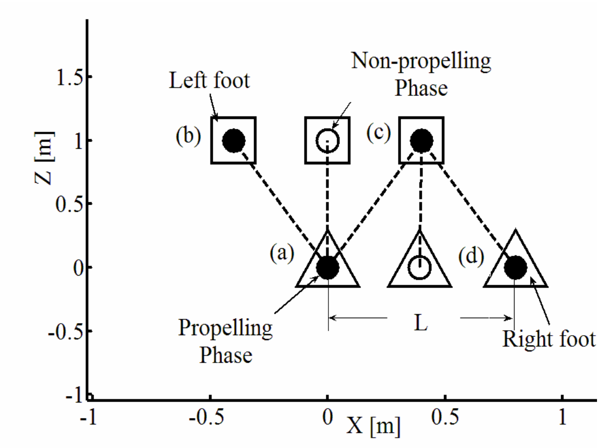


Fig. 19. The motion sequences of feet in horizontal plane when biped robot walks on the ground.

In Fig. 19, the triangles depict the right foot and the rectangles depict the left foot. The transition moment between a step is depicted with blank rectangles and triangles. Fig. 20 shows the correspondent logic flowchart of the walking gaits.

By varying the design parameters, a parametric study has been carried out. Some preliminary results are shown in Fig. 21 and Fig. 22. The motion trajectories for different parameters are depicted with dashed lines.

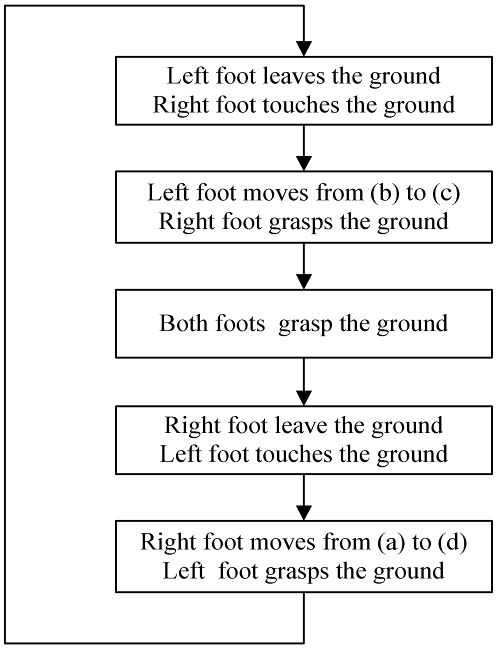


Fig. 20. The logic flowchart of walking gaits as in Fig. 19

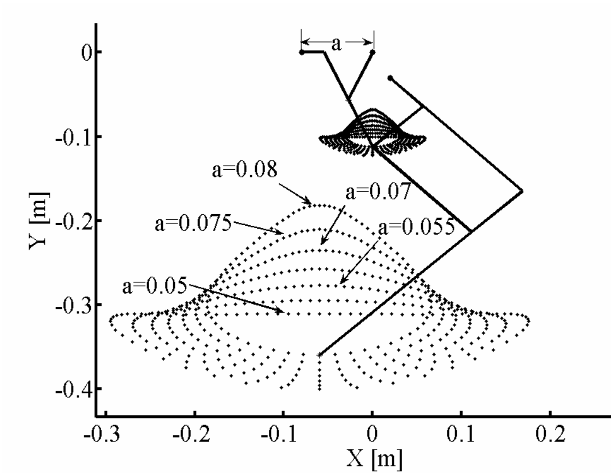


Fig. 21. The motion trajectories changes when the design parameter a is varying

It can be observed that parameter a will greatly affects the shape of the trajectory of Point A and parameter h affects the position of the trajectories. When increasing a, the length for a step will be shorten and step height will be enlarged and vice versa. The relationship between step length、step height and variable a、h seems to be linear. In addition, the ratio of the link size of Chebyshev linkage will also greatly affect the shape of trajectories.

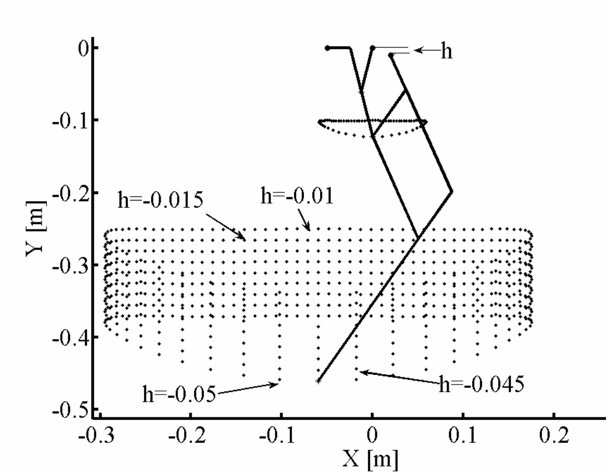


Fig. 22. The motion trajectories changes when the design parameter h is varying

5. Conclusions

In this paper, an operation analysis of a one DOF Chebyshev-pantograph leg mechanisms is carried out to characterize the design and operation. Kinematic equations of the proposed leg mechanism are formulated for a computed-oriented simulation in Matlab environment. Experiment results show that the prototype can performs biped walking actuated by only one DOF, but there are still some drawbacks. Simulation results show the feasible performances of proposed leg mechanisms. The relationship between design parameters and operation characteristics has been analyzed in a parametric study.

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